

Post-Quantum Mathematics

Part 3 - Standards Tracks & Candidates: Candidate Asymmetric Algorithms

Compiled by the Spaced Research Ledger

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This part covers the algorithms NIST did not choose. Some lost a competition. Some were never entered. Two have been standardized anyway, by other bodies, which is why they matter more than the word “candidate” suggests.

The reason to track them is concentration risk. The two algorithms most of the world is deploying, ML-KEM and ML-DSA, both rest on the difficulty of problems over structured lattices. If that structure turns out to be exploitable, the replacements will have to come from somewhere, and they will come from this part. Germany and France have already made that argument in policy, as Part 2 describes.

Three families appear below. Code-based schemes rest on the difficulty of decoding a random-looking error-correcting code, and are the oldest of the three by decades. Isogeny-based schemes rest on relationships between elliptic curves, and produce the smallest keys of any family. Lattice-based alternatives use the same broad mathematics as the NIST winners, but deliberately without the algebraic structure.

Section 1 covers BIKE, Section 2 Classic McEliece, and Section 3 isogeny schemes after the SIKE break. Section 4 covers the conservative lattice alternatives, and Section 5 proposals on no standardization track at all.

1. BIKE

BIKE reached NIST's final code-based round and lost to HQC. Its unresolved issue is that nobody can prove how often it fails to decrypt, which is not a normal thing to leave open in a deployed cipher.

Current State

BIKE is a code-based key encapsulation mechanism that went through NIST's fourth round alongside Classic McEliece and HQC, and was not the selection [1].

Its open problem is the decoding failure rate. That rate is currently estimated only by extrapolation, with no proven upper bound established, and this bears directly on its security claim against adaptive attacks [2]. A scheme that occasionally fails to decrypt can leak information through those failures, so an unproven bound is a real gap rather than a technicality.

A 2026 multi-instance analysis found a further weakness. For one BIKE parameter set, security drops below the target of 143 bits once an attacker targets $M \geq 2^8$ session keys at once [2]. This is a degradation under multiple simultaneous targets rather than a single-instance break, but the multi-target setting is the realistic one for anything protecting many connections.

Researchers have proposed standardizing two configurations going forward: a reduced-cost variant for forward-secrecy uses, or a version with a proven security guarantee obtained through an improved decoder [3].

2. Classic McEliece

Classic McEliece has resisted attack for roughly 45 years, was rejected by NIST, and was then standardized by ISO. Two papers published within three days of each other in August 2026 eroded its security margin without breaking it.

Recent Developments

A paper dated 24 August 2026 costs the best-known key-recovery attacks against all Classic McEliece parameter sets [14]. It reports conditional bit-operation estimates below the NIST classical reference level for each one, for example $2^{126.77}$ for the smallest parameter set. These are conditional asymptotic cost estimates, not demonstrated practical attacks.

A second paper, submitted 7 August 2026 and revised on the 27th, presents a classical distinguisher running in quasipolynomial time, $n^{O(\log n)}$ [6]. It applies to every Classic McEliece parameter set considered in the NIST process, in a particular asymptotic regime. The authors describe the result as yielding improved but not yet practical concrete attack estimates.

Two independent results narrowing the margin in the same month is worth noting, even though neither is a break.

Current State

Against that, the scheme's record is the strongest in this part. Its binary Goppa-code foundation has resisted practical attack across roughly 45 years of published cryptanalysis, and its NIST-considered parameter sets remain unbroken in practice [5].

ISO standardized Classic McEliece under its asymmetric ciphers standard, reported as of July 2026 [4]. That is a real standardization track for a scheme NIST declined to select, and it is why Germany can recommend it while the United States does not.

The practical drawback is unchanged. Its public keys are far larger than most competing schemes, and several compact variants proposed over the years to shrink them have subsequently been broken [5]. The large key is doing security work.

3. Isogeny-Based Schemes After SIKE

This family suffered the most dramatic failure in post-quantum cryptography, and did not go away. Work continues because its keys are far smaller than anything else on offer.

Current State

SIKE was broken in July 2022 by an attack exploiting auxiliary point information in the protocol specification, enabling classical key recovery in roughly one hour on standard hardware [7]. A NIST finalist falling to a laptop in an hour is the reference event this whole area is defined against.

Research continued regardless, driven by the compact key sizes the family offers, and multiple newer schemes are designed specifically to resist the attack that broke SIKE [7].

A June 2026 paper presents isogeny-based signatures with randomizable keys, part of a wider trend toward isogeny signatures using higher-dimensional constructions for practical efficiency [8]. Isogeny-based signatures remain among the most compact post-quantum signature constructions known. SQIsign, the one isogeny scheme still in NIST's process, is covered in Parts 1 and 5.

4. Conservative Lattice Alternatives

FrodoKEM uses the same broad mathematics as ML-KEM and deliberately throws away the structure that makes ML-KEM fast. Two European governments prefer it for exactly that reason.

Current State

FrodoKEM was standardized by ISO in June 2026, as an amendment to the asymmetric ciphers standard [9].

Its ISO track has a documented history. The relevant ISO working group agreed in April 2023 to move forward with standardization [10]. In March 2025 FrodoKEM was separately submitted to the IETF's Crypto Forum Research Group as an Internet-Draft.

The design choice is the whole argument. FrodoKEM's security rests on the Learning With Errors problem over generic, algebraically unstructured lattices, unlike ML-KEM's structured ones, specifically to minimize exposure to future attacks that exploit algebraic structure [9]. The cost is size and speed. The benefit is that an attack on structured lattices would not obviously apply.

One other alternative appears in the record. A scheme from the NTRU Prime family is optimized for constrained environments and was an alternative candidate in NIST's original process, though not selected [11]. No 2026-specific security or standardization update for it was found beyond that background.

5. Proposals on No Standardization Track

Not every post-quantum construction is trying to become a standard. These two are recorded as examples of what independent academic work is building on the standards-track primitives.

Current State

RingSLIP is a 2026 ring-signature construction built from the Lattice Isomorphism Problem, using HAWK as a component [12]. Its signature size grows only logarithmically with the number of members in the ring. It is worth noting that HAWK itself was withdrawn from NIST's process in July 2026, for reasons Part 5 covers.

LoTRS is a separate 2026 lattice-based threshold ring signature construction, also on no standardization track, aimed at practical two-round threshold signing [13].

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Sources

1. HQC and BIKE - https://asecuritysite.com/hqc_bike (undated)
2. Multi-instance security analysis of BIKE - <https://eprint.iacr.org/2026/517.pdf> (undated)
3. A Lean BIKE KEM - <https://csrc.nist.gov/csrc/media/Events/2024/fifth-pqc-standardization-conference/documents/papers/a-lean-bike-kem.pdf> (undated)
4. ISO - Classic McEliece (team page on ISO/IEC 18033-2 standardization) - <https://classic.mceliece.org/iso.html> (as of 2026-07-15)
5. Classic McEliece cryptanalytic history - <https://eprint.iacr.org/2026/1512> (undated)
6. Quasipolynomial Cryptanalysis of the McEliece Cryptosystem - <https://eprint.iacr.org/2026/1630> (as of 2026-08-27)
7. On the SIKE break and subsequent isogeny work - <https://eprint.iacr.org/2026/446.pdf> (undated)
8. Isogeny-based signatures with randomizable keys - <https://eprint.iacr.org/2026/1169> (undated)
9. FrodoKEM - <https://frodokem.org/> (as of 2026-06-01)
10. draft-longa-cfrg-frodokem-security-considerations - <https://datatracker.ietf.org/doc/html/draft-longa-cfrg-frodokem-security-considerations/> (as of 2025-03-01)
11. NTRU Prime in constrained environments - <https://arxiv.org/html/2508.10023v1> (undated)
12. RingSLIP: ring signatures from the Lattice Isomorphism Problem - <https://eprint.iacr.org/2026/889> (undated)
13. LoTRS: lattice-based structured threshold ring signatures - <https://eprint.iacr.org/2026/974> (undated)
14. Bit Operation Cost of Holdout Key-Recovery Attacks Against Classic McEliece - <https://eprint.iacr.org/2026/1786> (as of 2026-08-24)